

Ankit Kumar

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ABOUT

AI/ML Engineer and Backend Developer with expertise in building production-grade LLMs and RAG systems. Proven track record in translating complex data science into scalable software, from optimizing autonomous systems at ISRO via Reinforcement Learning to deploying high-impact conversational AI. Dedicated to developing efficient, robust solutions for AI/ML and software engineering challenges.

EXPERIENCE

Indian Institute of Technology, Guwahati | AI/ML Research

June 2026 – Present | Guwahati, Assam, India

- Leading research on next-gen LLM architectures, focusing on Transformer efficiency, reasoning, and scalability.
- Implementing deep learning models (PyTorch) to explore multimodal models and Mixture of Experts (MoE).

AI/ML Engineer | Freelance

July 2024 – June 2026 | Remote

- Deployed 4+ production RAG systems, achieving 20% improvement in retrieval performance.
- Collaborated with the University of Stuttgart to develop a material tensile analysis platform, automating complex stress-strain calculations for laboratory research.
- Trained 124M parameter LLM from scratch and scaled conversational AI to 5,000 daily interactions.

ISRO - Indian Space Research Organisation | Project Intern

May 2024 – June 2024 | Chennai, India

- Optimized Lunar landing via RL, reducing trajectory error by 15% for autonomous mission simulations.
- Engineered data pipelines processing 10k+ points/sec, enabling real-time fault forecasting with 90% accuracy.
- Doubled telemetry throughput via low-level Python/C layers, conditioning GBs of sensor data for RL training.

Ecowiser | Backend Developer Intern

Nov. 2022 - Apr. 2023 | Hyderabad, India

- Scaled Django-based backend for 10,000+ active users across Web and Chrome Extension platforms.
- Boosted internal search speed by 30% using custom NLP tokenization in PostgreSQL.
- Automated customer engagement with Redis/Celery, processing 5,000+ scheduled dispatches daily.

Velozity Global Solutions | Machine Learning Intern

June 2022 - Oct. 2022 | Chennai, India

- Developed PyTorch/TensorFlow frameworks for 15GB+ unstructured datasets.
- Improved temporal dependency identification by 14% using benchmarked LSTM architectures.

PROJECTS

Tensile Analysis Platform

Mar. 2026 – May 2026

- Designed and deployed a high-performance material characterization dashboard used by a University of Stuttgart research laboratory, automating 90% of complex stress-strain analysis.
- Built custom mathematical physics solvers, including an optimized Young's modulus window-search engine using cumulative sum statistics and a strict 0.2% yield strength offset script.
- Implemented fuzzy-token matching to integrate 100+ unstructured laboratory spreadsheets with raw data, embedding automated statistical analysis (ANOVA, t-tests, IQR outlier detection).

ModelSmith

June 2022 – July 2022

- Developed a unified model-evaluation web application in Django hosting 10+ distinct machine learning architectures, including custom ResNets and Generative Adversarial Networks (GANs).
- Authored thorough technical specifications and documentation detailing the underlying linear algebra, backpropagation mechanics, and production deployment parameters.

EDUCATION

Lovely Professional University

Bachelor of Technology in Computer Science and Data Science

Oct. 2020 – May 2024 | Punjab, India

- Relevant Coursework: Data Structures, Algorithms, Databases, Computer Systems, Machine Learning, Artificial Intelligence

CERTIFICATIONS

Reinforcement Learning (NPTEL)

Jan - Apr 2022 | NPTEL, IIT Madras

Python for Data Science (Elite)

Jan - Feb 2022 | NPTEL, IIT Madras

SKILLS

- Languages & DBs:** Python, SQL (PostgreSQL, MySQL), Ruby, C
- AI/ML & Frameworks:** PyTorch, TensorFlow, LLMs, RAG, Reinforcement Learning, LangChain, LlamaIndex, Django (DRF), Fastapi, Pandas, NumPy, Scikit-learn
- DevOps & Tools:** Docker, Git, Linux, CI/CD, Redis, Celery, Cloud Platforms